

Manual

For Remote Control



AutoWave

**Portable solution to measure,
simulate and analyse
Battery Supply Voltage Variation**



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1 Interfaces

1.1 Ethernet

10/100Base-T interface

10Mbit/sec or 100Mbit/sec

Ethernet Address selectable (default: 10.0.0.2)

Ethernet Gateway selectable (default: 10.0.0.1)

Ethernet Net mask selectable (default: 255.0.0.0)

1.1.1 Transmission Control Protocol (TCP)

TCP Port: 15000

1.1.2 FTP

FTP Port: 21

Username: guest

Password: no password

1.2 Parallel IEEE 488 interface

- Addresses 1 - 30 selectable
- Command: (SH1, AH1, T4, L2, SR1, RL2, PP1, DC0, DT0, C0, E1)
- Connector and pin layout as per IEEE - 488 - 1975
- 24-pin Amphenol connector
- 8 ground pins

1.3 Timing

- It is recommended to use a minimum delay of 250ms between two commands

1.4 Firmware

These commands are implemented in Firmware version 5.09.07 or higher

2 Parameters

	Name	Min – Max	Step	Unit	Delivery value
Polarity	Pol.	0: UNIPOLAR 1: BIPOLAR	-	-	0, 1
Input Voltage DC Source	InDC	1 - 10	1	V	1 - 10
Output Voltage DC Source	OutDC	1 - 999	1	V	1 - 999
Measuring Input Voltage	MeasIn	5, 10, 20, 50, 100	-	V	5, 10, 20, 50, 100
Filename Segments	SegFile	*.dsg, *.tsg	-	-	*.dsg, *.tsg
Filename Points	PntFile	*.dpt, *.tpt	-	-	*.dpt, *.tpt
Filename Record	RecFile	*.rec, *.trc	-	-	*.rec, *.trc
Filename PFM200N	SwitchFile	*.dsw, *.tsw	-	-	*.dsw, *.tsw
Events	Evnt	0: File Events, -1: endless, 1 - 9'999'999	1	-	-1, 0, 1 - 9'999'999
ON /OFF	ON/OFF	ON, OFF	-	-	ON, OFF
Voltage	Voltage	-100 - +100		V	-100.0 - +100.0 (float)
Pathname	Path	-	-	-	Pathname
Filename	File	-	-	-	Filename
DUT Monitor Action	Action	Disable, Notify, Stop	-	-	0, 1, 3
Triggers	TrigGen	Off Manual @ start Trigger In @ start Automatic Manual @ event Trigger In @ event Manual @ iteration Trigger In @ iteration	-	-	0 1 2 3 4 5 6 7
	TrigOutx	Off Start Stop Event Break Iteration	-	-	0 1 2 3 4 5
	TrigInx	Off On	-	-	0 1
Directory Pathname	DirPath	Download Dir. Record Dir. Upgrade Dir. Log Files Dir.			DOWD, RECD, UPGD, LOGD
Date	Timestamp	Unix time, number of seconds since 00:00:00 UTC on January 1, 1970	1	s	0 – 2147483647

3 Commands

COMMUNICATION		
Comm.	Syntax	Description
Identification	*IDN?	The identification code includes the model name, the coding of hardware options, and the version of the firmware's. Answer: <manufacturer>, <model>, <0>, <Firmware level>, <Outputs>, <Inputs> *IDN:EM TEST, AutoWave, 0, 5.06.02, 4, 2
Reset	*RST	Stops the communication
Go to local	*GTL	Stops a running test and returns to local mode
Echo	*ECHO:<ON/OFF> *ECHO?	Switch echo on / off Ask for echo state
Reboot	REB	Reboot the machine
License	LCN <key> LCN?	License Answer: <OK> / <ERR>
Protocol	*PRCL:<ON/OFF> *PRCL?	Switch protocol on / off Ask for protocol state

TEST SETUP				
Comm.			Syntax	Description
Range	RANG	OUT1	RANG:OUT1 < Pol InDC OutDC >	Sets the In/Output range of the DC Source
		OUT2	RANG:OUT1 < Pol InDC OutDC >	
		OUT3	RANG:OUT1 < Pol InDC OutDC >	
		OUT4	RANG:OUT1 < Pol InDC OutDC >	
		IN1	RANG:IN1 < MeasIn>	
		IN2	RANG:IN2 < MeasIn>	
Mode	MOD	GEN	MOD:GEN REC GNRC	Set to Mode Generator Set to Mode Recorder Set to Mode Gen. & Rec.
Source	SOUR	SEGM	SOUR:SEGM <SegFile>	Select the play file
		POIN	SOUR:POIN <PntFile>	
		POIN	SOUR:POIN <RecFile>	
Events	EVNT		EVNT <Evnt>	Set the number of events
Set Voltage	VSET	OUT1-4	VSET:OUT1 <Voltage>	Set the output voltage
Set Offset	VOFS	OUT1-4	VOFS:OUT1 <Voltage>	Set the output offset
Trigger	TRIG	GEN	TRIG:GEN <TrigGen>	Set the start trigger (play)
		OUT1-2	TRIG:OUT1 <TrigOutx> TRIG:OUT2 <TrigOutx>	Set the "trigger OUTx" property
		IN2	TRIG:IN2 <TrigInx>	Set the "trigger IN2" property

TEST				
Comm.			Syntax	Description
START	STAR		STAR	Start
STOP	STOP		STOP	Stop
BREAK	BREA		BREA	Break
DUT Monitor	DUTM	IN1, IN2	DUTM:IN1 <Action>	Set the DUT Monitor mode
	DUTM?	IN1, IN2	DUTM?:IN1	Ask the DUT Monitor mode

FILES				
Comm.			Syntax	Description
SIZE	SIZ?		SIZ? <path>	Ask for file size
TRANSMIT	TRFL		TRFL <path>	Initialise a file download. If :ERR, filename already exists on target.
	TRFL?		TRFL? <path>	Initialise a file upload. If :ERR filename doesn't exists on target
DELETE	DEL		DEL <path>	Delete a file on target. If :ERR, file doesn't exist
DIR?	DIR?		DIR? <DirPath>	Get the paths of DOWD, RECD, UPGD, LOGD or content of <pathname>
CHECK	CKFL?		CKFL? <path>	Check if file downloaded
	CKLF?		CKLF? <Filename>	Ask for duration, channels, events, trigger and master channel of a test file
	CKHD?		CKFL? <path.dpt>	Save header from <path.dpt> under </home/guest/LogFiles/header.hpt>
FLNM?	FLNM?		FLNM? DUTM	Get the path and filename of file which contains all DUT Events
			FLNM? ERR	Get the path and filename of file which contains all process errors

Not classified				
Comm.			Syntax	Description
DISPLAY	DISP		DISP <string>	Display a 40 characters long string (lowest line)
DATE	DAT		DAT <timestamp>	Set the date
	DAT?		DAT?	Ask for the date (returns the actual timestamp)

STATE				
Comm.			Syntax	Description
STATE TEST	STAT?	TEST	STAT? TEST <i>answer: <testval></i>	State of all channels
STATE IN	STAT?	IN1-2	STAT? IN1 <i>answer: <inval></i>	State of a chosen input
STATE OUT	STAT?	OUT1-4	STAT? OUT1 <i>answer: <outval></i>	State of a chosen output
STATE SYST	STAT?	SYST	STAT? SYST <i>answer: <sysval></i>	System Versions
STATE BATT	STAT?	BATT	STAT? BATT <i>answer: <battval></i>	Battery state
STATE MAC	STAT?	MAC	STAT? MAC <i>Answer: <macval></i>	MAC Address
STATE DUTM	STAT?	DUTM	STAT? DUTM <i>Answer: <dutval></i>	DUT Monitors
STATE ERR	STAT?	ERR	STAT? ERR <i>answer: <errval></i>	Get error code /cleared after read
STATE DUTM IN	STAT:D UTM:	IN1-2?	STAT:DUTM:IN1? <i>Answer:<dutlevel></i>	Get DUT Monitor level
STATE DLTM	STAT?	DLTM	STAT? DLTM <i>Answer:<segments>,<points>,<iterations></i>	Get the minimal latency time between segments, points and iterations

4 Feedback Messages

Message	Description	
OK	The command has been accepted and will be executed	
ERR	The command is not a valid command	
testval	Out 1 <statval> Out 2 <statval> Out 3 <statval> Out 4 <statval> In 1 <statval> In 2 <statval> Dut <dutstat>	State Test Value STAT TEST:2,0,0,2,3,4,0
inval	In <statval> Dut <dutstat> Time <time>	State In Value STAT IN1:2,0,0,0.02
outval	Out <statval> Dut <dutstat> Iteration Total, Iteration Current, Event, SegNr, Time <time>, Test time	State Out Value STAT OUT1:6,0,1,1,0,2,0.28,4.35
sysval	HW AutoWave, FW AutoWave, NAME FB, HW Version FB, FW Version FB, HW DSP, FW DSP, SN DSP, Calibration Date, FB Unique ID	System Value SYST:FWV_AW,1.20t55;NAME_FB,AUTOWAVE_FRAME BOARD;HWV_FB,101039- 2;FWV_FB,0.53a01;HWV_DSP,101066- 0;FWV_DSP,1.07.00;SN_DSP,00000000;CAL,25072006;UI D_FB,00:00:00:00:00:5E
statval	0: stopped 1: ready 2: started 3: fail 4: fast 5: DSP not ok 6: break 7: not ready 8: finished 9: iterate 10: undefined 11: writetofile 13: file processing	Status Value
battval	0: nc 1: ok 2: low 3: empty	Current battery state
dutstat	0: no dut monitor event 1: dut monitor event occurred	DUT Monitor Status
dutval	0: nothing 1: notify 3: stop	DUT Monitor Value STAT DUTM:1
dutlevel	0 : inactive 1 : active	

Message	Description	
errval	0: No Error 1: Internal Error 2: Disk Space 3: Record Time 4: Source Polarity 5: Record sample rate to high 6: Number of channels 7: Scale factor 8: Sample rate 9: Send Sequence to DSP 10: Acquiring Rate 11: Calibration Data 12: DUT Monitor 13: Play file name 17: Memory 18: Firmware not able to play 23: File error 24: File reading error 26: File processing error 27: Not supported segment 28: SegmentTrigger not supported 29: PointsTrigger not supported 30: OutTrigger configuration conflict 31: Unsupported OutTrigger configuration 33: Segment checking fails 50: Module error 51: Module update in progress 52: Can't play with module	Error Value
macval	MAC Address STAT MAC: 00:E0:4B:10:F7:FB	
Time	Time in seconds	

5 Protocol

The communication between the host computer and the AutoWave could be more secure by implementing a protocol.

The protocol could be switched ON/OFF by the command “ *PRCL:<ON/OFF> “ At switch on of equipment the protocol is OFF.

All commands with first character * has no protocol.

For any request a response should come within a preset time (i.e. should be less than 300 ms, which is normally the timeout for a disconnection detection).

Answer can be one of the following:

Simple answers (when no decoration is expected):

- ACK: acknowledge: command is understood and treated.
- NAK: not acknowledge: command is not understood or checksum is wrong.
- NOTREADY: command cannot be accepted (various reasons: busy with another command, ...)
- BUSY: command is understood, but treatment is (already) in progress).

Decorated answers:

- A decorated response that is always equivalent to an ACK.

If a BUSY answer comes, the remote caller is expected to send again the previous message (the called device cannot spontaneously answer when the treatment of a command is achieved. It is only expected to send an answer when asked.)

If the remote caller sends a new command while a previous is still in treatment, then the caller will be notified with a NOTREADY (this command was not the expected one) and the current treated command will be flushed as soon as possible. (Remark: any further call will return NOTREADY until the previous command is effectively flushed.)

5.1 Message format

The message starts with a control mark STX followed by the command. At least the control mark ETX and the checksum

	STX	Command	ETX	CS
Characters	1	X	1	1

Command: The characters are IBM-PC characters from 20H to FFH.

CS: Checksum. For checksum value less than or equal to 1FH, 20H must be added.

5.1.1 Control element of the message:

The control marks in the messages are special characters as follows:
(ASCII character from 00H to 1FH)

Elements		Hex-code	Comment
STX	Start of TeXt	02H	Begin Calc. Checksum.
ETX	End of TeXt	03H	End Calc. Checksum.
ACK	ACKnowledge	06H	Command is understood and treated
NACK	Negative ACKnowledge	15H	command is not understood or checksum is wrong
BUSY	Busy	19H	command is understood, but treatment is (already) in progress
NOTREADY	Not ready	16H	command can not be accepted (various reasons: busy with another command, ...)

Rem: BUSY uses the regular End of Medium (EM) character control, NOTREADY uses the regular Synchronous Idle (SYN) character control.

5.1.2 Calculation of Checksum

The checksum will be calculated by adding the ASCII codes between the STX and ETX control and appends to the end of the message.

If the checksum is less or equal to 0x20, 0x20 is added again. Thus ensures that the checksum is not interpreted as control character.

STAT? PSRC	Command		Checksum	
	Decimal	Hex	Decimal	Hex
S	83	0x53	83	0x53
T	84	0x54	167	0xA7
A	65	0x41	232	0xE8
T	84	0x54	316	0x13C
?	63	0x3F	379	0x17B
	32	0x20	411	0x19B
P	80	0x50	491	0x1EB
S	83	0x53	574	0x23E
R	82	0x52	656	0x290
C	67	0x43	723	0x2D3
0x00FF & Checksum			211	0xD3

LCN?	Command		Checksum	
	Decimal	Hex	Decimal	Hex
L	76	0x4C	76	0x4C
C	67	0x43	143	0x8F
N	78	0x4E	221	0xDD
?	63	0x3F	284	0x11C
0x00FF & Checksum			28	0x1C
Checksum < 0x20			60	0x3C

STX	Command				ETX	CS
0x02	0x4C	0x43	0x4E	0x3F	0x03	0x3C

5.1.3 Example of message

STX	STAT? PSRC										ETX	CS
0x02	0x53	0x54	0x41	0x54	0x3f	0x20	0x50	0x53	0x52	0x43	0x03	0xD3

STX	LCN?				ETX	CS
0x02	0x4C	0x43	0x4E	0x3F	0x03	0x3C

5.1.4 Example of answers

Remote: <STX>CKSR? file.dsg<ETX><CS>

Answer sample 1: <STX>CKSR file.dsg: 300, 20000, 20000, 0.102400;<ETX><CS>

Answer sample 2: <STX>CKSR file.dsg: ERR;<ETX><CS>

Answer sample 3: <BUSY>

Answer sample 4: <NOTREADY>

Answer sample 5: <NAK>

6 Examples

6.1 Initialization

	Command	Answer
Read Identification	*IDN?	*IDN:EM TEST, AutoWave, 0, 5.09.00, 4, 2
Switch echo on	*ECHO:ON	*ECHO ON:OK
Switch protocol on	*PRCL ON	*PRCL ON:OK
Read System version	STAT? SYST	STAT SYST: FWV_AW,5.09.00; NAME_FB,AUTOWAVE_FRAMEBOARD; HWV_FB,101039-2; FWV_FB,0.60a01; HWV_DSP,101066-0; FWV_DSP,3.31.00; SN_DSP,0000000; CAL,01012003; UID_FB,00:00:00:00:00:5E
Read MAC address	STAT? MAC	STAT MAC: 00:E0:4B:25:AA:F2
Get the minimal latency time	STAT? DLTM	STAT DLTM: 0.000000, 0.180000, 0.070000
Sets the In/Output range of the DC Source	RANG OUT1-4 ,0,10,100	RANG OUT1-4,0,10,100
Ask for available modules	GTMD?	GTMD:<Empty if no module connected>
Set the output voltage	VOFS:OUT1-4 0	VOFS:OUT1-4 0
Set the output offset	VSET:OUT1-4 13.5	VSET:OUT1-4 13.5
License check	LCN?	LCN:xxxxx-xxxxx-xxxxx-xxxxxxxxxxxx

6.2 Start a test file

	Command	Answer
Read download path	DIR? DOWD	DIR DOWD:/home/guest/DowFiles
Initialize transfer	TRFL /home/guest/DowFiles/SineTest.dsg	TRFL /home/guest/DowFiles/SineTest.dsg:OK
Set mode to Generate	MOD GEN	MOD GEN
Select file to generate	SOUR SEGM SineTest.dsg	SOUR SEGM SineTest.dsg
Start	STAR	STAR
Read status	STAT? OUT1	STAT OUT1:13,0,0,0,0,0,0,0.00,-1
	STAT? OUT1	STAT OUT1:2,0,1,1,0,0,9.94,0.06,-1
	STAT? OUT1	STAT OUT1:8,0,1,1,0,0,0.00,10.00,-1

* The file *SineTest.dsg* is generated by the autowave.control software.

To stop the generator

	Command	Answer
Start the test	See above	
Read status	STAT? OUT1	STAT OUT1:2,0,1,1,0,0,9.94,0.06,-1
Stop the test	STOP	STOP
Reset the output offset	VSET:OUT1-4 13.5	VSET:OUT1-4 13.5

7 Modules commands

7.1 Modules Identification and Status

Ask available modules	GTMD?	GTMD? answer: <id>,<id>,...	Ask for available modules (id = address of module)
Ask upgrade progress	UPGD: STAT?	MODL:UPGD:STAT? answer: <id>,<model>,<FWV>,<% of progress>	Used in case of "STAT? ERR 51" module update in progress
Ask module identification	IDN?	MODL:IDN? <id> answer: <id>,<manufacturer>,<model>,<0>,<firmware version>,<type>[,<othersinfo>]	Ask for module identification <othersinfo> is specific to module
Ask module system	SYST?	MODL:SYST? <id> answer: <id>,<name>,<fwv>,<hwv>,<uid>,<blv>	Ask for module system details
Status	STAT?	MODL:STAT? <id> answer: <id>,<modlstatus>	Ask for the module status
Error	ERR?	MODL:ERR? <id> answer: <id>,<modlerror>[,<errorinfo>]	Ask for the module error, modlerror and errorinfo are specific to module
	RST	MODL:ERR:RST <id>	Reset the last module error
Limits	LIM?	MODL:LIM? <id> answer: <id>,<limits>	<limits> is specific to module

Parameter	Name	Value / Min-Max	Unit	Comment
Identifier	id	050100 : default for AMP 200N 060100: default for PFM 200N 070100: default for VDS 200Q 080100: default for SNG 200P		Address of modules
System	name	Name of module		
	fwv	Firmware version		
	hwv	Hardware version		
	uid	Unique identifier		
	blv	Bootloader version		
Module status	modlstatus	0 : Ready 1 : Module disconnected 2 : Module error 3 : test ON (button Off)	4 : Wait for Error Acquit 5 : Restarting 6 : Busy 10: Undefined	

7.2 Specific for AMP 200Nx

Parameter	Name	Value / Min-Max	Unit	Comment
Other info	othersinfo	<Has Measure>,<Amplifier type>,<Amplifier FWV>,<has DC mode>		
Error	modlerror	0 : No Error 1 : Busy (1) 2 : Test On State (1) 3 : Fail (2) 4 : Over temperature (3) 5 : Overload (3) 6 : Safety Loop (3) 7 : Master Absent (3)		(1) Already defined in Module Status (2) Need a module hard reset (3) Module restart automatically after command MODL:ERR:RST
	errorinfo	<powerdisp>,<temp>		
Measure option	Has Measure	0 : No measure option 1 : Measure option installed		
Amplifier type	Amplifier type	<Power in W>.<Revision>		Ex. 800.3
Amplifier FWV	Amplifier FWV	Firmware version of amplifier		
DC mode option	has DC mode	0 : No DC mode 1 : DC mode option installed		
Power dissipation	powerdisp		W	

Temperature	temp		°C	
Limits	limits	<fmin>,<fmax>,<sgout Vmin>,<sgout Vmax>,<2W Vmin>,<2W Vmax>,<2W Imin>,<2W Imax>		f: device frequency [Hz] sgout V: signal out voltage [V] 2W V: 2wire/coax out voltage [V] 2W I: 2wire/coax out current [A]

Module AMP 200Nx Setup

Range	RANG	MODL:RANG <id>,<moduleOut>,<pol>,<+Vin>,<+Vout>	Set the range of module output
Input setup	SETUP	MODL:SETUP:IN1 <id>,<masterOut> MODL:SETUP:IN2 <id>,<masterOut>	
DC source mode	DC	MODL:SETUP:DC <id>,<OnOff>	Set DC source mode On or Off
	DC?	MODL:SETUP:DC? <id> answer: <id>,<OnOff>	Usable only if DC mode option is installed
Frequency range	EXRNG	MODL:SETUP:EXRNG <id>	Set DDS frequency max. to 500kHz

Parameter	Name	Value / Min-Max	Unit	Comment
Polarity	pol	0 : Unipolar 1 : Bipolar		
In / out voltage	+Vin / +Vout		V	Positive max values
Master output	masterOUT	-1 : none 0 : CH1 1 : CH2 2 : CH3 3 : CH4		
DC source mode On or OFF	OnOff	0 : OFF 1 : ON		

AMP 200Nx Signal output setup

Subset	Signal	SGNL:	
Signal path	PATH	MODL:SGNL:PATH <id>,<in1out>,<in2out>,<DDSout>	Set the signal path
	PATH?	MODL:SGNL:PATH? <id> answer: <id>,<in1out>,<in2out>,<DDSout>	Ask for the signal path
Source Switch AMP 200N2 only	SRCE	MODL:SGNL:SRCE <id>,<OnOff>	Set the Source Switch state
	SRCE?	MODL:SGNL:SRCE? <id> answer: <id>,<OnOff>	Ask for the Source Switch state
Voltage range	VRNG	MODL:SGNL:VRNG <id>,<vrange>	
Amplifier range	LFAR	MODL:SGNL:LFAR <id>,<lfarange>	Set LFA range
	LFAR?	MODL:SGNL:LFAR? <id> answer: <id>,<lfarange>	Ask for LFA range

Parameter	Name	Value / Min-Max	Unit	Comment
Module output	in1out / in2out / DDSout	0 : None 1 : Signal Out 2 : 2Wire 3 : Coaxial		
Voltage range	vrange	0 .. Vmax	Vp	(float)
Amplifier range	lfarange	0 : Low range 1 : High range 2 : Middle range		

AMP 200Nx Signal

Signal setting	DATA	MODL:SGNL:DATA <id>,<sigValues>	Set the signal values
Signal offset	OFST	MODL:SGNL:OFST <id>,<level>,<rampfactor>	Set the signal offset
	OFST?	MODL:SGNL:OFST? <id> answer: <id>,<level>	Ask for the signal offset
Limits	LIM?	MODL:SGNL:LIM[ALL]? <id>	Fmin,Fmax,Fstp,Vmin,Vmax,Vstp,dBmin,dBmax,dBstp,Imin,Imax,Istp
Signal start	STAR	MODL:SGNL:STAR <id>	
Signal stop	STOP	MODL:SGNL:STOP <id>,<end>	

Signal status	STAT?	MODL:SGNL:STAT? <id>	Ask for the Signal Status
		answer: <id>,<sigstatus>,<dutstatus>,<powerdisp>,<currstresttime>,<elapstottime>	

Parameter	Name	Value / Min-Max	Unit	Comment
Signal Values	sigValues	<frequency>,<amplitude>,<unit>,<dweltime>,<breake>,<rampfactor>		
	frequency	0 ... 250000 [500000]	Hz	
	amplitude	0 ...10.0	Vp	
		-60 ... +40	dB	
	unit	V , dB		
	dweltime	0.1 ... 9999	s	
	breake	0 ... 60	s	
	rampfactor	<= 0 : No ramp 1.. 1000000000	uVp/ms	
Signal stop state	end	0 : Stop 1 : maintains signal		
Signal Status	sigstatus	0 : Stopped 1 : Started 2 : Broken 3 : Finished 4 : Maintained 5 : Fail 6 : Working 10 : Undefined		
DUT Monitor Status	dutstatus	0 : No DUT Monitor event 1 : DUT Monitor event occurred		
Rest Time	currstresttime		s	
Total Time	elapstottime		s	

AMP 200Nx Measure

Subset	Measure	MEAS:	
Measure Source	SRCE	MODL:MEAS:SRCE <id>,<measIn>	Set the Measure Source
	SRCE?	MODL:MEAS:SRCE? <id> answer: <id>,<measIn>	Ask for the Measure source
Read Measure	READ?	MODL:MEAS:READ? <id>[,<frequency>] answer: <id>,<frequency>,<value>,<unit>	If <frequency> isn't specified, use signal frequency

Parameter	Name	Value / Min-Max	Unit	Comment
Measure input	measIn	1 : Voltage 2 : Current Clamp		
Measure Values	frequency	10 – 250000	Hz	
	value		'V'	

AMP 200Nx Parameters

Subset	Param	PARM	
Power dissipation	POWD	MODL:PARM:POWD? <id>	
		answer: <id>,<value>	
Temperature	TEMP	MODL:PARM:TEMP? <id>	
		answer: <id>,<value>	

AMP 200Nx Calibration

Subset	Calib	CAL	
Calibration status	STAT?	MODL:CAL:STAT? <id>	0 : Error (not calibrated)
		answer: <id>,<Status>	1 : Calibration OK

7.3 Specific for PFM 200Nx

Parameter	Name	Value / Min-Max	Unit	Comment
Other info	othersinfo	<has idle setting>		
Error	modlerror	0 : No Error 1 : Busy (1) 2 : Test ON (button Off) (1) 3 : Over temperature (3) 4 : Overload (3) 5 : Memory full 6 : Segment not OK (4) 7 : OverRun error (3) 8 : Hard error (5) 9 : Sequence error (3) 10 : ReversOverCurrent (3) 11 : OverVoltage 50 : Other error (2)		(1) Already defined in Module Status (2) Need a module hard reset (3) Module restart automatically after command <i>MODL:ERR:RST</i> (4) Segment is not compatible with switch setting (5) Internal Hardware error, need EMTEST service.
	errorinfo	Void for PFM200		
Limits	limits	<power Vmax>,<power Imax>,<data Vmax>,<data Imax>,<off time min>,<on time min>,<cycle time min>,<safe time min>		Vmax in Volts Imax in Amps Time min in μ s
Idle setting	Has idle setting	0 : No idle setting (switch closed with TestON) 1 : Idle state can be defined		

Module PFM 200Nx Setup

Data Idle	IDLE	MODL:SETUP:IDLE <id>,<DataLines>,<PowerLines>
-----------	------	---

Parameter	Name	Value / Min-Max	Unit	Comment
Idle status	status	0 : Switch open 1 : Switch closed		Set the idle state for all switch

7.4 Specific for VDS 200Qx

Parameter	Name	Value / Min-Max	Unit	Comment
Status	othersinfo	Void for VDS 200Q		
Error	modlerror	0 : No Error 1 : Busy 2 : Test ON (button Off) 3 : Over temperature 4 : Over voltage 5 : Current limitation (Info) 6 : Error power fail 7 : Error safety fail 8 : Segment Error 9 : Power source not started 10 : Error amplifier not detected 11 : Error amplifier power supply 20 : Other error		Already defined in Module Status Need a module hard reset
	errorinfo	Void for VDS 200Q		
Limits	limits	<Vmin>,<Vmax>,<Imax>,<lpeak>,<peak Time>,<fmax>		V in Volts, I in Amps, f in Hz, time in ms

Module VDS 200Qx Capability

Subset	Capability	CAP:	
Voltage range	VRNG?	MODL:CAP:VRNG? <id> answer: <id>,<available ranges>	Capabilities are detailed below for each device.
Amplifier gain	GAIN?	MODL:CAP:GAIN? <id> answer: <id>,<available gains>	

Frequency Compensation	FCMP?	MODL:CAP:FCMP? <id>	
		answer: <id>,<available compensations>	
Current limiter	IMAX?	MODL:CAP:IMAX? <id>	
		answer: <id>,< available current limit mode>	

Information: Capability for each device.

- VDS 200Q25, Q50, Q100 , Q150 , Q200 (and .1):
 - Voltage range: 1 = low range (Vmax = 30V)
2 = full range (Vmax = 60/80V)
 - Amplifier gain: 1 = low gain (x1)
2 = High gain (x7)
 - Current limiter mode: 0 = disable (current limitation is done by the hardware)
1 = peak OFF (no inrush current)
2 = lcontrol (inrush of 3x programmed current limit for 200ms)
3 = lmax (inrush of 3x maximum current of device for 200ms)
 - Frequency Compensation: 1 = standard (bw ≈ 40kHz)
2 = capacitive (bw ≈ 3kHz)
3 = high frequency (bw ≈ 150/250kHz)
- VDS 200Q10:
 - Voltage range: 1 = 2 = fix to full range (Vmax = 60V)
 - Amplifier gain: 1 = low gain (x1)
2 = High gain (x7)
 - Current limiter mode: 1 = fix to peak OFF (no inrush current)
 - Frequency Compensation: 1 = standard (bw ≈ 40kHz)
2 = capacitive (bw ≈ 3kHz)
3 = high frequency (bw ≈ 180kHz)
- VDS200Qx.2:
 - Voltage range: 1 = fix
 - Amplifier gain: 1 = low gain (x4)
2 = High gain (x8)
 - Current limiter mode: 0 = disable (current limitation is done by the hardware)
1 = peak OFF (no inrush current)
2 = 3*I (inrush of 3x programmed current limit for 200ms)
3 = 3*I_{max} (inrush of 3x maximum current of device for 200ms)
 - Frequency Compensation: 1 = standard (bw ≈ 40kHz)
2 = capacitive (bw ≈ 3kHz)
3 = high frequency (bw ≈ 150/250kHz)

Module VDS 200Q Setup

Subset	Setup	SETUP:	
Device mode	MOD	MODL:SETUP:MOD <id>,<devMode>	
	MOD?	MODL:SETUP:MOD? <id> answer: <id>,< devMode >	
Input setup	IN	MODL:SETUP:IN1 <id>,<masterOut>	
Source setup	SRCE	MODL:SETUP:SRCE<id>,<vRange>,<gain>,<iLimitMode>,<fComp>	Setup the source parameters
	SRCE?	MODL:SETUP:SRCE? <id> answer: <id>,<vRange>,<gain>,<iLimitMode>,<fComp>	
Current limit	CURRENT	MODL:SETUP:CURRENT <id>,<lmaxValue>	
Voltage limit	VLIM	MODL:SETUP:VLIM <id>,<VmaxPos>,<VmaxNeg>	VDS 200Qx.2 Only
	VLIM?	MODL:SETUP:VLIM? <id> answer: <id>,< VmaxPos>,<VmaxNeg>	
Output Impedance	OIMP	MODL:SETUP:OIMP <id>,<ImpValue>	VDS 200Qx.2 Only
	OIMP?	MODL:SETUP:OIMP? <id> answer: <id>,< ImpValue>	

Parameter	Name	Value / Min-Max	Unit	Comment
Device mode	devMode	0: FrameBus Off (VDS stand alone or other master) 1: FrameBus On (AutoWave is master) 2: Running mode (AutoWave is master and signal goes out)		

Master output	masterOUT	-1 : none 0 : CH1 1 : CH2	2 : CH3 3 : CH4	ex.: 2 = Channel 3 of AutoWave connected to Analogue input of VDS	
Output range	vRange	See §7.4 Module Capability, Information: Capability for each device			
Amplifier gain	gain				
Current limitation mode	iLimitMode				
Frequency Compensation	fComp				
Current limitation	ImaxValue	0 - Imax		A	
Voltage limitation	VmaxPos VmaxNeg	Pos : 0 – device Vmax Neg : Device Vmin – 0		V	
Output Impedance	ImpValue	OFF(0) / 10 – 200 Step : 10 mΩ		mΩ	OFF = Amplifier impedance

7.5 Specific for SNG 200P

Parameter	Name	Value / Min-Max	Unit	Comment
Other info	othersinfo	Void for SNG 200P		
Error	modlerror	0 : No Error 1 : Busy 2 : Test ON (button Off) 6 : Download error 50 : Other error		
	errorinfo	Void for SNG 200P		
Limits	limits	<Vmax>,<off time min>,<on time min>,<cycle time min>,<safe time min>		Vmax in Volts Time min in μs

Module SNG 200P Sequences

Save sequences	SAV	MODL:SAV <id>,<SequenceID>,<File>
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Parameter	Name	Value / Min-Max	Unit	Comment
Sequence ID	SequenceID	1 - 4		
File to save	File	"Name.tsw"		Temporary switched wave Default: SNG_SEQUENCE_X.TSW